

## MIT Academic Departments and Programs

MIT has 32 academic departments and programs spread across its five schools and the Schwarzman College of Computing. Each department is identified by a course number used in the MIT curriculum.

### School of Architecture and Planning

Department of Architecture, Course 4. The Department of Architecture offers programs in architecture, computation, and design. Research areas include building technology, history and theory, and design computation. Degrees offered are Bachelor of Science in Art and Design, Master of Architecture, SMArchS, and Doctor of Philosophy.

Department of Urban Studies and Planning, Course 11. This department focuses on city and regional planning, housing, transportation, environmental policy, and community development. Degrees offered are Bachelor of Science, Master in City Planning, and Doctor of Philosophy.

Program in Art Culture and Technology. An interdisciplinary graduate program for artists and designers exploring the intersections of technology, culture, and society. Degree offered is Master of Science.

MIT Media Lab and Media Arts and Sciences. The Media Lab is a world-famous research laboratory investigating the future of human-computer interaction, wearable technology, synthetic biology, and creativity. Degrees offered are Master of Science and Doctor of Philosophy.

### School of Engineering

Aeronautics and Astronautics, Course 16. This department covers fluid dynamics, astrodynamics, autonomous systems, propulsion, and space systems engineering. Degrees offered are Bachelor of Science, Master of Science, Doctor of Philosophy, and Doctor of Science.

Biological Engineering, Course 20. This department integrates biology and engineering to develop new materials, medicines, and bioprocesses. Degrees offered are Bachelor of Science, Master of Science, and Doctor of Philosophy.

Chemical Engineering, Course 10. Research areas include process systems, energy, nanotechnology, and computational chemistry. Degrees offered are Bachelor of Science, Master of Science in Chemical Engineering, and Doctor of Philosophy.

Civil and Environmental Engineering, Course 1. This department covers structural engineering, geotechnics, environmental fluid mechanics, water resources, and transportation. Degrees offered are Bachelor of Science, Master of Science, Doctor of Philosophy, and Doctor of Science.

Electrical Engineering and Computer Science, Course 6. This is MIT's largest department. Research areas include artificial intelligence, machine learning, circuits, communications, robotics, systems, and quantum computing. Undergraduate tracks are 6-1 for Electrical Science and Engineering, 6-2 for Electrical Engineering and Computer Science combined, 6-3 for Computer Science and Engineering,

and 6-7 for Computer Science and Molecular Biology. Graduate degrees include Master of Engineering, Master of Science, and Doctor of Philosophy.

Materials Science and Engineering, Course 3. This department covers metals, polymers, semiconductors, and biomaterials with a focus on quantum materials and sustainable energy. Degrees offered are Bachelor of Science, Master of Science, Doctor of Philosophy, and Doctor of Science.

Mechanical Engineering, Course 2. Research spans thermodynamics, fluid mechanics, design, robotics, and manufacturing. Degrees offered are Bachelor of Science, Master of Science, Naval Engineer degree, Doctor of Philosophy, and Doctor of Science.

Nuclear Science and Engineering, Course 22. Research areas include plasma physics, fusion energy, nuclear reactor design, and radiation biology. Degrees offered are Bachelor of Science, Master of Science, Doctor of Philosophy, and Doctor of Science.

Institute for Data Systems and Society. This is an interdisciplinary institute bridging statistics, machine learning, social science, and engineering systems. The doctoral degree offered is the PhD in Social and Engineering Systems.

## **School of Humanities Arts and Social Sciences**

Economics, Course 14. This highly ranked program focuses on microeconomics, macroeconomics, development economics, and public finance. Degrees offered are Bachelor of Science and Doctor of Philosophy.

Global Studies and Languages, Course 21G. This department offers language and culture courses in Chinese, French, German, Japanese, Portuguese, Russian, Spanish, and comparative media studies.

History, Course 21H. Covers American, European, Asian, and world history with a particular focus on the history of science and technology. Degrees offered are Bachelor of Science and Doctor of Philosophy.

Linguistics and Philosophy, Course 24. Research areas include formal semantics, syntax, philosophy of mind, and logic. Degrees offered are Bachelor of Science and Doctor of Philosophy.

Literature, Course 21L. Spans British, American, and world literature with coursework in creative writing and film studies.

Music and Theater Arts, Course 21M. Offers performance ensembles, composition, and music technology courses alongside theater production.

Political Science, Course 17. Research focuses on international relations, security studies, comparative politics, and American politics. Degrees offered are Bachelor of Science and Doctor of Philosophy.

Science Technology and Society, Course STS. An interdisciplinary program examining the social and ethical dimensions of science and technology. Degrees offered are Bachelor of Science and Doctor of

Philosophy.

Writing Rhetoric and Professional Communication, Course 21W. Provides instruction in scientific writing, technical communication, and creative writing.

### **MIT Sloan School of Management**

Sloan School of Management, Course 15. MIT Sloan is a top-ranked business school with specializations in finance, entrepreneurship, operations, analytics, and sustainability. Degrees offered include Bachelor of Science in Management Science, Master of Business Administration, Executive MBA, Master of Finance, Master of Business Analytics, and Doctor of Philosophy.

Sloan Fellows Program. A one-year management program for mid-career professionals from around the world leading to a Master of Science in Management.

### **School of Science**

Biology, Course 7. Covers molecular biology, genetics, biochemistry, and structural biology. Research is closely tied to the Whitehead Institute and the Broad Institute. Degrees offered are Bachelor of Science and Doctor of Philosophy.

Brain and Cognitive Sciences, Course 9. Examines perception, learning, memory, and neural computation. Research spans neuroscience, psychology, and artificial intelligence. Degrees offered are Bachelor of Science and Doctor of Philosophy.

Chemistry, Course 5. Research spans organic, inorganic, physical, and biological chemistry. Degrees offered are Bachelor of Science, Doctor of Philosophy, and Doctor of Science.

Earth Atmospheric and Planetary Sciences, Course 12. Covers geology, oceanography, atmospheric science, and planetary science. Degrees offered are Bachelor of Science, Master of Science, Doctor of Philosophy, and Doctor of Science.

Mathematics, Course 18. Covers pure and applied mathematics including algebra, analysis, combinatorics, and computational science. Degrees offered are Bachelor of Science, Master of Science, and Doctor of Philosophy.

Physics, Course 8. Research areas include condensed matter physics, high-energy physics, astrophysics, and quantum information. Degrees offered are Bachelor of Science, Doctor of Philosophy, and Doctor of Science.

### **Schwarzman College of Computing**

Artificial Intelligence and Decision-Making, Course 6-4. An undergraduate major combining computer science, artificial intelligence, and social science to train students who can build and critically evaluate

AI systems.

Computer Science, Course 6-3 and Course 18C. Core computer science education spanning algorithms, programming languages, systems, and theory of computation.

Statistics and Data Science Center. An institute-wide hub coordinating research and education in statistics, machine learning, and data science. Offers an interdisciplinary doctoral minor.

### **Key Research Laboratories and Centers**

MIT Lincoln Laboratory is a federally funded research and development center focusing on national security technology.

The Computer Science and Artificial Intelligence Laboratory, known as CSAIL, is the world's largest university AI and computer science research lab.

The Koch Institute for Integrative Cancer Research merges biology and engineering for cancer research.

The Broad Institute is a joint MIT and Harvard genomics and biomedical research center.

The MIT Energy Initiative, known as MITEI, supports cross-disciplinary research on sustainable energy.

The Picower Institute for Learning and Memory conducts neuroscience research on learning, memory, and cognition.

The Research Laboratory of Electronics, known as RLE, focuses on fundamental research in electronics, photonics, and quantum systems.

The Plasma Science and Fusion Center, known as PSFC, is home to the SPARC compact fusion reactor project.